

# Amdahl's Law

## 1. Objective

The objective of the program is to show the effect of Amdahl's Law through experimental comparison of execution time for an unoptimized algorithm and an optimized algorithm. Unoptimized algorithm – Bubble Sort, Optimized algorithm - Merge Sort. The experimental overall speedup will be compared with theoretical speedup based on Amdahl's law.

## 2. Implementation

The program creates a data set consisting of 30,000 pseudorandom integers. For each data set a random seed is fixed at 12345 before generation so that the algorithms Bubble sort and Merge sort work with the same set of integers.

Program consists of two parts:

- Serial part: creation of a random data set in memory.
- Processing part: sorting the created data.

Unoptimized algorithm – Bubble Sort, Optimized algorithm - Merge Sort.

The program measures the total time of execution and processing only time of execution for each algorithm.

## 3. Environment and Execution

**Language:** C

**Compiler:** GCC (GNU Compiler Collection)

**Dataset Size:** 30,000 integers

**Random Seed:** 12345

The program was compiled and executed using the following commands:

```
gcc -std=c11 -O0 sort_performance.c -o lab1
./lab1
```

The -std=c11 option specifies the C11 language standard. The -O0 option disables compiler optimization so that the experiment primarily compares the algorithmic difference between Bubble Sort and Merge Sort.

## 4. Experimental Results

The following results were obtained:

| Measurement              | Result           |
|--------------------------|------------------|
| T_unoptimized            | 1.619403 seconds |
| T_processing_unoptimized | 1.619015 seconds |

| Measurement                 | Result           |
|-----------------------------|------------------|
| T_optimized                 | 0.003862 seconds |
| T_processing_optimized      | 0.003437 seconds |
| Measured Overall Speedup    | 419.317193       |
| P                           | 0.999760         |
| S                           | 471.054699       |
| Theoretical Overall Speedup | 423.373333       |

Both algorithms successfully sorted their datasets, as verified by the program's correctness check.

## 5. Amdahl's Law Analysis

```
=====
CSE 3320 Lab 1 - Amdahl's Law
Bubble Sort vs. Merge Sort
=====
Dataset size: 30000 integers
Random seed: 12345

UNOPTIMIZED VERSION (Bubble Sort)
-----
T_unoptimized = 3.693330 seconds
T_processing_unoptimized = 3.692329 seconds

OPTIMIZED VERSION (Merge Sort)
-----
T_optimized = 0.008295 seconds
T_processing_optimized = 0.007412 seconds

AMDahl's LAW RESULTS
-----
Measured Overall Speedup = 445.247740
P = 0.999729
S = 498.155559
Theoretical Overall Speedup = 439.002734

CORRECTNESS CHECK
-----
Bubble Sort result: Sorted correctly
Merge Sort result: Sorted correctly
root@Anil:/mnt/c/Users/012an/OneDrive - UT Arlington/Desktop/VS code#
```

### Measured Overall Speedup

The measured overall speedup is calculated using:

$$\text{Measured Speedup} = T_{\text{unoptimized}} / T_{\text{optimized}}$$

Using the measured values:

$$\text{Measured Speedup} = 1.619403 / 0.003862$$

$$\text{Measured Speedup} = 419.317193$$

Therefore, the optimized implementation was approximately **419.32 times faster overall** than the unoptimized implementation.

### Proportion Improved (P)

P represents the proportion of the original execution time spent in the portion of the program that was improved.

$$P = T_{\text{processing\_unoptimized}} / T_{\text{unoptimized}}$$

$$P = 1.619015 / 1.619403$$

$$P = 0.999760$$

Therefore, approximately **99.976%** of the original execution time was spent in the processing portion that was improved.

### Processing Speedup (S)

S represents the speedup achieved for the improved portion of the program.

$$S = T_{\text{processing\_unoptimized}} / T_{\text{processing\_optimized}}$$

$$S = 1.619015 / 0.003437$$

$$S = 471.054699$$

Therefore, the optimized processing portion was approximately **471.05 times faster** than the unoptimized processing portion.

### Theoretical Overall Speedup

Amdahl's Law gives the theoretical overall speedup as:

$$\text{Theoretical Speedup} = 1 / ((1 - P) + (P / S))$$

Using the measured P and S values:

$$\text{Theoretical Speedup} = 423.373333$$

The theoretical speedup predicted by Amdahl's Law is therefore approximately **423.37 times**.

## 6. Comparison and Discussion

The actual overall speedup obtained was 419.32×, whereas Amdahl's Law indicated a theoretically possible speedup of 423.37×. The two results are very similar, indicating that the actual speedup value is consistent with the theoretical calculation.

The discrepancy between the two results may occur due to various factors that influence program execution, such as the accuracy of timing, the effects of the operating system, memory allocation, etc.

Moreover, the described experiment illustrates the core idea of Amdahl's Law, which states that an improvement in a particular part of a program could dramatically decrease the execution time, but the execution time of the non-improved part would limit this improvement. In this particular

experiment, most of the time was used for Bubble Sort, which explains the considerable speedup by switching to the more efficient sorting algorithm.

## 7. Conclusion

In this case, Amdahl's Law was proven using Bubble Sort and Merge Sort algorithms operating on the same set of data. Bubble Sort algorithm took about 1.619 seconds of CPU time, while Merge Sort took about 0.00344 seconds. Speedup ratio found for this test was 419.32 times faster than Bubble Sort, while Amdahl's Law suggests it should be 423.37 times.

The high degree of accuracy proves that Amdahl's Law is a great tool for calculating overall program improvement ratio when only a part of a program is optimized.